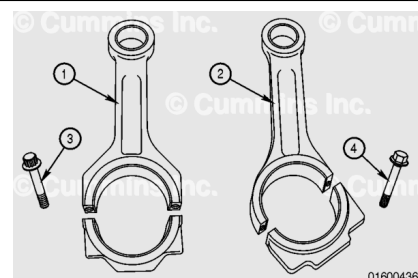


Trainings ▾

General Information

The QSK45 and QSK60 engines utilize two connecting rod designs. They are a straight split connecting rod (1) and an angle split connecting rod (2).

To identify a connecting rod installed in an engine, check the connecting rod capscrew. The straight split connecting rod capscrew (1) has a double hexagonal, 12 point, head (3). The angle split connecting rod capscrew (2) has a single hexagonal, 6 point, head (4).



Engines that contain the straight split connecting rod require the connecting rod, piston, and liner to be removed as a complete assembly. Refer to Procedure 001-045 in Section 1.

(</qs3/pubsys2/xml/en/procedures/56/56-001-045-tr.html>)

Use the following procedure for removal of the angle split connecting rods. Refer to Procedure 001-054 in Section 1.

(</qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html>)

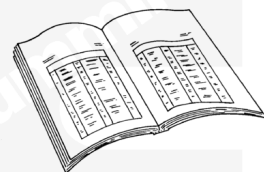
Note : Angle split connecting rods from fracturing rigs, excavators, loaders and mechanically driven haul trucks, **must not** be reused at rebuild or repaired, regardless of hours. Angle split connecting rods installed in QSK45/60 engines in electrically driven haul trucks, power generation, marine or other applications **must only** be reused at rebuild or repair if the connecting rods have less than 10,000 hours and pass the reuse guidelines. If a connecting rod does not meet the reuse guidelines, it **must not** be reused, regardless of hours. Straight split connecting rods from any application can be reused regardless of the hours, provided they pass the reuse guidelines.

Note : Angle split and straight split connecting rods **can not** be mixed within an engine. All connecting rods **must** be the same design. Straight split connecting rods **can not** be installed into engines built prior to Engine Serial Number (ESN) 33172060, unless they have had a cylinder block upgrade.

Preparatory Steps

- Remove the piston from the connecting rod. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html>)

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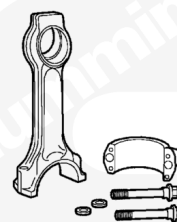
ck800wa

Clean and Inspect for Reuse

⚠ CAUTION ⚠

The number on the connecting rod must be the same as the number on the cap. **NEVER** assemble a new cap to an old rod or an old cap to a new rod.

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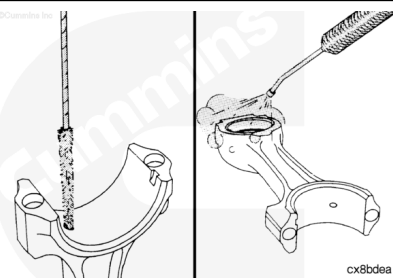
cx400ga

Remove the capscrews and the cap from the connecting rod.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

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cx8bdea

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam to clean the parts.

Dry with compressed air.

Use a soft bristle brush to clean the oil drilling.

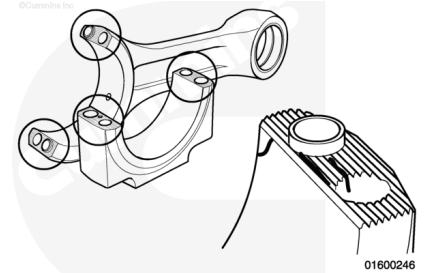
Dry the rod with compressed air.

Inspect the rod and the cap for fretting damage.

The rod and cap **must** be replaced, as an assembly, if fretting damage is visible on either piece.

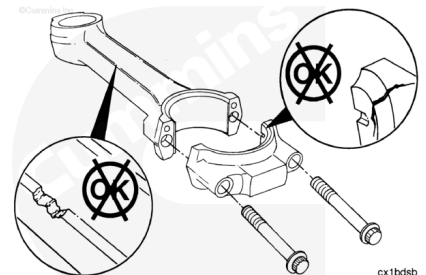
Inspect for cracks and serration around the oil drilling.

Note : The straight split connecting rod does **not** have a serrated joint, as seen on the angle split connecting rod.



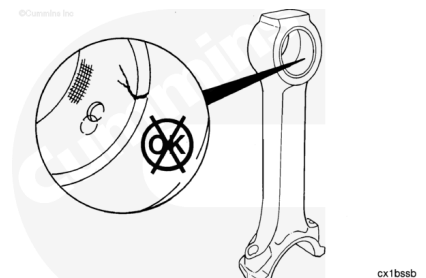
Inspect the rods and caps for damage.

Replace the rod and cap, if either the I beam or cap is nicked or otherwise damaged.

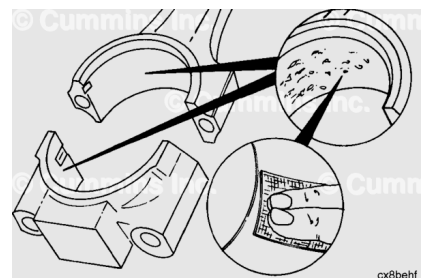


Inspect the rod pin bore bushing for damage or misalignment of the oil hole passage and bushing.

If the bushing is damaged, the rod **must** be replaced.

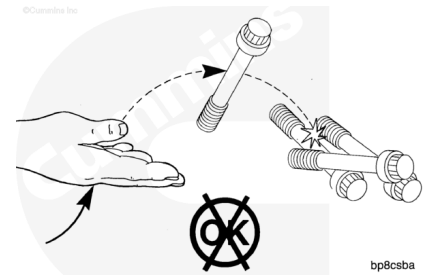


Inspect the bearing surface for nicks or burrs. If it is **not** possible to remove any nicks or burrs with a fine crocus cloth, the rod **must** be replaced.



⚠ CAUTION ⚠

Prevent damage to the capscrews. Nicks in the body of the capscrew can cause an area of stress that can fail during engine operation. Damage to the threads will cause torque values to be incorrect and will damage the mating parts.



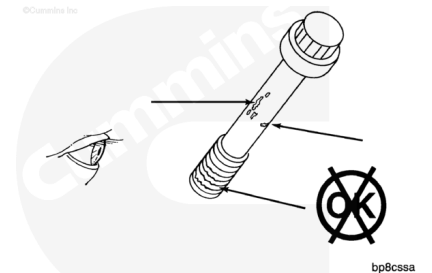
Note : Repair of rolled threads by the use of a thread die is **not** recommended. The thread die can create a sharp corner on the minor diameter (root) of the threads. This sharp corner can cause an area of increased stress.

Inspect the capscrew.

The capscrew **must** be replaced if:

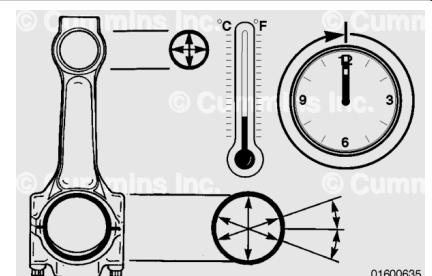
- The threads are damaged
- Rust or corrosion has caused pitting in the body
- The body is nicked, galled, bent, or stretched.

Perform a magnetic crack inspection of the connecting rod capscrews. Connecting rod capscrews **must** be magnetically inspected for cracks. Refer to Procedure 017-001 in Section 17. (</qs3/pubsys2/xml/en/procedures/56/56-017-001-tr.html>)



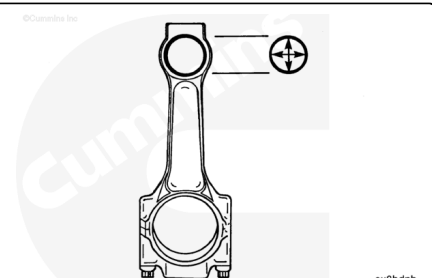
Measure

Measurement of the connecting rod bores requires measuring equipment with a high level of precision and **must only** be taken after the connecting rods temperatures have stabilized for 24 hours at a stable temperature between 21 to 24°C [70 to 75°F].



Measure the connecting rod bushing inside diameter.

Connecting Rod Bushing Inside Diameter - New		
mm		in
65.035	MIN	2.5604
65.047	MAX	2.5609



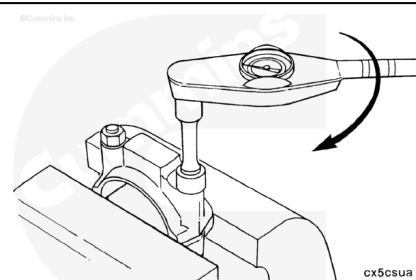
Connecting Rod Bushing Inside Diameter - Used		
mm		in
65.025	MIN	2.5600
65.060	MAX	2.5614

Note : The connecting rod bushing deforms during operation and may become out of round, resulting in the difference in new and used dimensions.

Note : The bushing is **not** replaceable. If the bushing is **not** within specification, the connecting rod **must** be replaced.

⚠ CAUTION ⚠

Use a vise with brass jaws to hold the connecting rod. Notches, scratches, or dents in the I-beam area can lead to engine failure.



⚠ CAUTION ⚠

The number on the connecting rod must be the same number as the number on the rod cap. NEVER assemble a new cap to an old rod or an old cap to a new rod.

Lubricate the capscrew head flange with clean gear oil (SAE W140).

Lubricate the capscrew threads with clean engine oil (SAE 15W40).

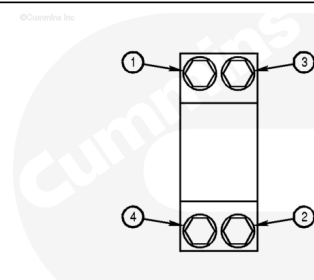
Assemble the connecting rod, cap, and capscrews.

Tighten the capscrews alternately and evenly to pull the cap over the dowel pins. Use the following steps and sequence to tighten the capscrews.

Angle Split Connecting Rod Torque Sequence:

Torque Value:

1. 85 n•m [63 ft-lb]
2. Repeat Step 1
3. Loosen all capscrews 2 revolutions minimum
4. 85 n•m [63 ft-lb]
5. Repeat Step 4



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6. Advance 60 degrees.

Straight Split Connecting Rod Torque Sequence:

Torque Value:

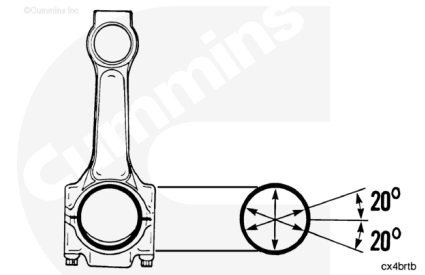
1. 115 n•m [85 ft-lb]
2. Repeat Step 1
3. Loosen all capscrews 2 revolutions minimum
4. 95 n•m [70 ft-lb]
5. Repeat Step 4
6. Advance 60 degrees.

Use a dial bore indicator. Measure the inside diameter using the measuring positions shown.

Connecting Rod Bearing Bore Inside Diameter

Angle Split Connecting Rods		
mm		in
127.032	MIN	5.0013
127.083	MAX	5.0032

Straight Split Connecting Rods		
mm		in
127.032	MIN	5.0013
127.072	MAX	5.0028



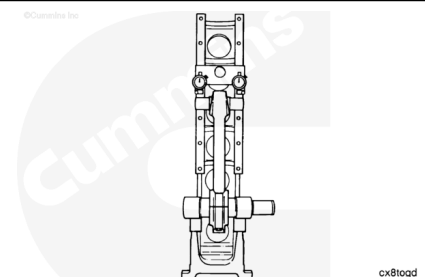
Note : If the measurements are not within specification, the connecting rod must be replaced.

Bend and Twist Inspect

Required tools:

- Part Number ST-561 Connecting Rod Checking Fixture
- Part Number 3163700 Connecting Rod Location Mandrel (for crank pin end of rod)
- Part Number 3163701 Connecting Rod Locating Mandrel (for piston rod end with bushing installed)

Bulletin 3377541, Service Tool Instructions - Connecting Rod Checking, is also available.



Calibration Procedure

⚠ CAUTION ⚠

The rod must be installed and torqued to specification or the measurement will be incorrect.

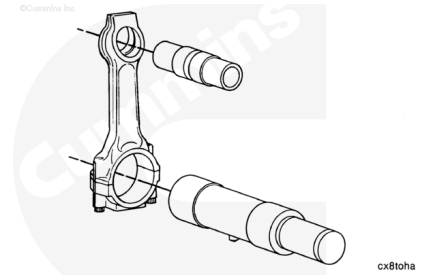
A connecting rod of known length, bend, and twist, called a master rod, is required to calibrate the fixture. A new connecting rod with a known distance from the center of the crank pin to the center of the piston pin end (rod length), can also be used.

Install the appropriate mandrel in the piston pin end of the master connecting rod.

Note : Be sure that the locating pin in the mandrel for the crank pin end is touching the connecting rod cap directly opposite the beam of the connecting rod.

Install the mandrel in the crank pin bore of the master rod.

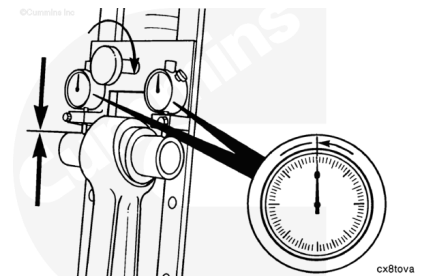
Tighten the mandrel so that it is centered in the bore correctly.



Install the master connecting rod in the cranking fixture.

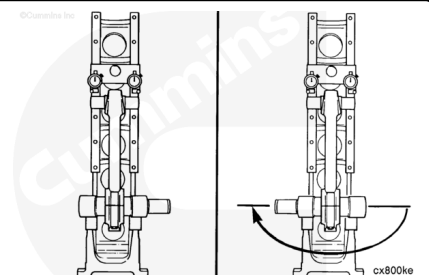
Loosen the knob and move the indicator bracket until both indicators touch the top of the piston pin end mandrel. Move the bracket toward the mandrel until the indicator needles have moved approximately 0.25 mm (0.010 in). Turn the knob to tighten the bracket.

Adjust the indicator needles to "0". Move the master connecting rod in and out to confirm the "0" setting.



Remove the master connecting rod from the fixture.

Rotate the connecting rod 180 degrees horizontally and install it in the checking fixture.



⚠ CAUTION ⚠

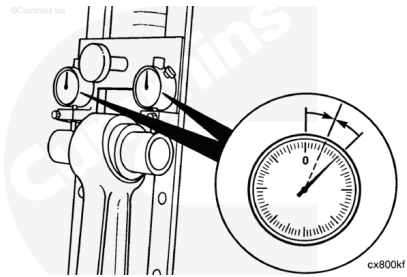
Do not adjust the indicators during this step of the procedure, or the measurement will be incorrect.

Be sure the indicators are at the "0" setting.

If the indicator needles do **not** return to the "0", adjust the indicator so that the "0" moves $\frac{1}{2}$ of the indicated difference from the needle to the "0" established during the previous step.

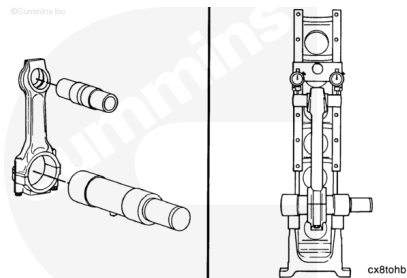
Note : If the needle is more than 0.10 mm [0.004 in] from the "0" established during the previous step, check for dirt or burrs on the mandrels and fixture. If the "0" is still **not** within specification, check to be sure that the master connecting rod is **not** damaged.

After completing the calibration of the fixture, remove the mandrels from the master rod.



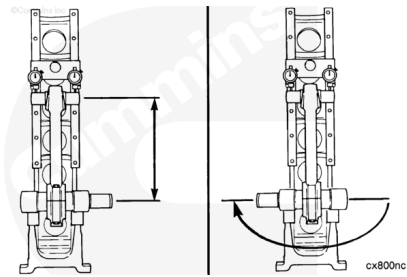
Repeat the same steps to install the mandrels in the connecting rod to be measured, as you did for the master rod.

Install the rod in the fixture.



Record the indicator reading. The difference of the calibrated "0" of the indicator **must** be added or subtracted from the known length of the master rod to determine the length of the connecting rod being measured.

Connecting Rod Length		
mm		in
320.937	MIN	12.635
320.987	MAX	12.637

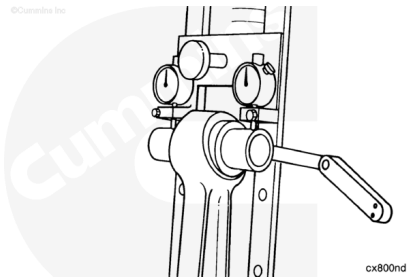


If the connecting rod length is **not** within specification, the connecting rod **must** be replaced.

Make a note of the indicator readings.

Remove the rod from the fixture. Turn the rod 180 degrees horizontally.

Compare the indicator readings with those noted in the previous step. The difference between the indicator readings is the amount of bend in the connecting rod.



⚠ CAUTION ⚠

Never attempt to straighten a connecting rod with the use of heat or force. The connecting rod will eventually break and cause extensive engine damage.

Maximum Connecting Rod Bend (Bushing Installed)

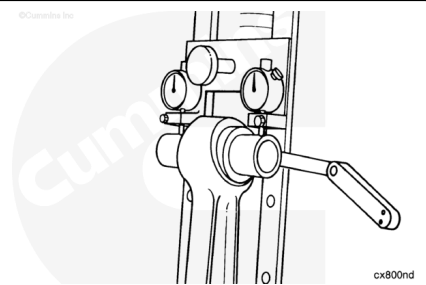
mm		in
0.10	MAX	0.004

If the connecting rod bend is **not** within specification, the connecting rod **must** be replaced.

Check the fixture and mandrel in the piston pin end for a gap between the two. If there is any twist in the connecting rod, the mandrel will **only** touch one side of the fixture.

Hold the end of the mandrel that is touching the fixture firmly against the fixture.

Use a feeler gauge to measure the gap between the mandrel and the fixture. The amount of gap between the mandrel and the fixture is the amount of connecting rod twist.



⚠ CAUTION ⚠

Never attempt to straighten a connecting rod with the use of heat or force. The connecting rod will eventually break and cause extensive engine damage.

Maximum Connecting Rod Twist (Bushing Installed)

mm		in
0.25	MAX	0.010

If the connecting rod twist is **not** within specification, the connecting rod **must** be replaced.

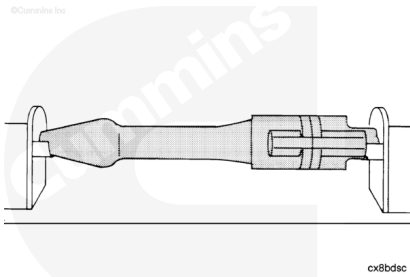
Magnetic Crack Inspect

Use a magnetic particle testing machine.

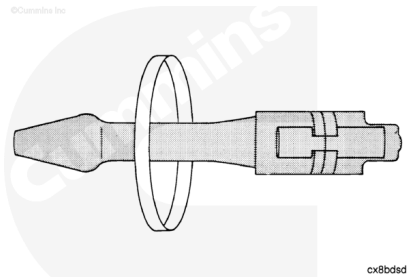
The connecting rod and cap **must** be assembled during this check.

Use the residual method. Apply head shot amperage. Adjust the amperage to 1500 ampere D.C. or rectified A.C.

Check for cracks.



Use the residual method. Apply coil shot amperage. Check for cracks.

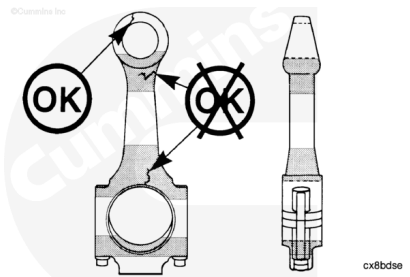


Amperage (Ampere Turns)	
MIN	2600 D.C. or Rectified A.C.
MAX	2800 D.C. or Rectified A.C.

Note : Ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

⚠ CAUTION ⚠

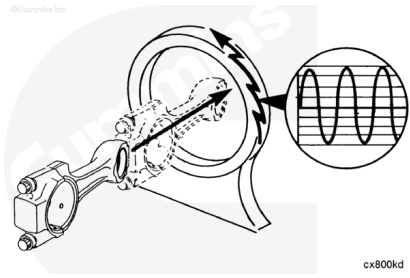
The rod must be demagnetized completely and cleaned thoroughly. Any small metal particles will cause engine damage.



The connecting rod **must** be replaced if any indications are visible in the critical (shaded) areas.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

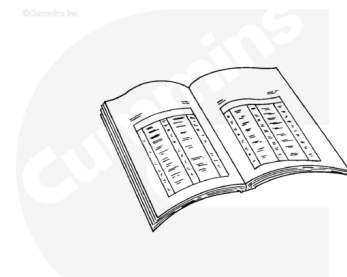
Demagnetize the connecting rod.

Use solvent or steam to clean the part.

Finishing Steps

- Install the piston onto the connecting rod. Refer to Procedure 001-054 in Section 1.
(</qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html>)

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